HW #5 — Random Variables: Expectations (25 pts)

MATH 241 — Probability, Fall 2026

Due Date: Friday, 10/16 at 11:59pm ET

Instructions

- Please submit this assignment to Gradescope as one .pdf files.
- Responses can be hand-written or typed.
- Certain questions will ask you to do the problem in a specific way. Please adhere to these instructions or points will be deducted.
- For all questions involving manual calculations, please show all work and round answers to four decimal places.
- Refer to syllabus for late work deductions.

Background

We will continue investigating more metrics that we can calculate with random variables, namely expectations and variances. An optional mid-semester survey is also included.

Part I: Textbook Problems (4 pts each)

From *Introduction to Probability, Second Edition*:

- Chapter 4: Q3, Q5, Q9

Part II: Other Problems

For each of the random variables listed below, I want you to find:

- The support of the random variable (i.e. $x \in X(s)$)
- Probabilities for each value of the support (i.e. $P(X = x)$)
- $E(X)$
- $E(X^2)$
- $Var(X)$ via $E(X)$ and $E(X^2)$
  - If common: do not use already-established expressions for $Var(X)$

These are the random variables to investigate:

1. $X \sim Bin(n = 2, p = 0.9)$
2. $X \sim f_X(x)$, where

$$f_X(x) = \begin{cases} \frac{1}{16} & \text{if } x = 0 \\ \frac{1}{2^x} & \text{for } x = 1, 2, 3, 4 \end{cases}$$

3. $X \sim Poisson(\lambda)$

Optional: Part III: Mid-Semester Survey

Can you believe it? We're halfway through the semester! As we head into fall break, I value feedback from you all regarding what has worked for you in this course and what has not.

Fill out the survey here: https://forms.gle/fLB9WkR4bbRtYeiH7

A Message

No homework will be assigned over fall break! Enjoy the break! :)